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Project Title	PAYMENT TRANSACTION MANAGEMENT SYSTEM
Week	Week 5

WEEK-4 TASK

Task Name

Payment Transaction Management System

TASK DESCRIPTION

1. Create Customer and Payment tables.
2. Store customer information and payment transaction details.
3. Track completed, pending, and rejected payments.
4. Compare payment methods and transaction values.
5. Prepare customer-wise and date-wise payment reports.

SQL CODE

```
CREATE DATABASE
```

```
CREATE DATABASE PaymentTrackingDB;
```

```
USE PaymentTrackingDB;
```

```
-- =====
```

```
-- 1. CREATE CUSTOMER TABLE
```

```
-- =====
```

```
CREATE TABLE Customer (
```

```
    customer_id INT PRIMARY KEY,
```

```
    customer_name VARCHAR(100) NOT NULL,
```

```

    email VARCHAR(100),
    phone VARCHAR(15)
);

-- =====

-- 2. CREATE PAYMENT TABLE

-- =====

CREATE TABLE Payment (
    payment_id INT PRIMARY KEY,
    customer_id INT,
    order_id INT,
    payment_mode VARCHAR(30) NOT NULL,
    payment_date DATE NOT NULL,
    amount DECIMAL(10,2) NOT NULL,
    payment_status VARCHAR(20) NOT NULL,

    FOREIGN KEY (customer_id)
    REFERENCES Customer(customer_id)
);

-- =====

-- 3. INSERT CUSTOMER DETAILS

-- =====

INSERT INTO Customer VALUES
(1, 'Rahul Kumar', 'rahul@gmail.com', '9100002001'),
(2, 'Priya Devi', 'priya@gmail.com', '9100002002'),
(3, 'Sanjay Raj', 'sanjay@gmail.com', '9100002003'),

```

```
(4, 'Keerthana Sri', 'keerthana@gmail.com', '9100002004'),  
(5, 'Vignesh Kumar', 'vignesh@gmail.com', '9100002005');
```

```
-- =====
```

```
-- 4. INSERT PAYMENT TRANSACTIONS
```

```
-- =====
```

```
INSERT INTO Payment VALUES
```

```
(701, 1, 3001, 'UPI', '2026-09-01', 8500.00, 'Successful'),  
(702, 2, 3002, 'Debit Card', '2026-09-01', 14200.00, 'Successful'),  
(703, 3, 3003, 'Net Banking', '2026-09-02', 5600.00, 'Pending'),  
(704, 4, 3004, 'Credit Card', '2026-09-02', 11750.00, 'Failed'),  
(705, 5, 3005, 'UPI', '2026-09-03', 9300.00, 'Successful'),  
(706, 1, 3006, 'Cash', '2026-09-03', 2500.00, 'Successful'),  
(707, 3, 3007, 'Debit Card', '2026-09-04', 6800.00, 'Failed'),  
(708, 2, 3008, 'UPI', '2026-09-04', 15600.00, 'Successful');
```

```
-- =====
```

```
-- 5. DISPLAY ALL CUSTOMERS
```

```
-- =====
```

```
SELECT * FROM Customer;
```

```
-- =====
```

```
-- 6. DISPLAY ALL PAYMENTS
```

```
-- =====
```

```
SELECT * FROM Payment;
```

```
-- =====
```

```
-- 7. DISPLAY SUCCESSFUL PAYMENTS
```

```
-- =====
```

```
SELECT *
```

```
FROM Payment
```

```
WHERE payment_status = 'Successful';
```

```
-- =====
```

```
-- 8. DISPLAY FAILED PAYMENTS
```

```
-- =====
```

```
SELECT *
```

```
FROM Payment
```

```
WHERE payment_status = 'Failed';
```

```
-- =====
```

```
-- 9. DISPLAY PENDING PAYMENTS
```

```
-- =====
```

```
SELECT *
```

```
FROM Payment
```

```
WHERE payment_status = 'Pending';
```

```
-- =====
```

```
-- 10. UPDATE PENDING PAYMENT
```

```
-- =====
```

```
UPDATE Payment
```

```
SET payment_status = 'Successful'
```

```
WHERE payment_id = 703;
```

```
-- =====
```

```
-- 11. PAYMENT DETAILS WITH CUSTOMER NAME
```

```
-- =====
```

```
SELECT
```

```
    p.payment_id,
```

```
    c.customer_name,
```

```
    p.order_id,
```

```
    p.payment_mode,
```

```
    p.payment_date,
```

```
    p.amount,
```

```
    p.payment_status
```

```
FROM Payment p
```

```
JOIN Customer c
```

```
ON p.customer_id = c.customer_id
```

```
ORDER BY p.payment_date;
```

```
-- =====
```

```
-- 12. COUNT TRANSACTIONS BY PAYMENT MODE
```

```
-- =====
```

```
SELECT
```

```
    payment_mode,
```

```
    COUNT(*) AS total_transactions
```

```
FROM Payment
```

```
GROUP BY payment_mode
```

```
ORDER BY total_transactions DESC;
```

```
-- =====
```

```
-- 13. TOTAL SUCCESSFUL AMOUNT BY PAYMENT MODE
```

```
-- =====
```

```
SELECT
```

```
    payment_mode,
```

```
    SUM(amount) AS total_amount
```

```
FROM Payment
```

```
WHERE payment_status = 'Successful'
```

```
GROUP BY payment_mode
```

```
ORDER BY total_amount DESC;
```

```
-- =====
```

```
-- 14. TRANSACTION COUNT BY STATUS
```

```
-- =====
```

```
SELECT
```

```
    payment_status,
```

```
    COUNT(*) AS total_transactions
```

```
FROM Payment
```

```
GROUP BY payment_status;
```

```
-- =====
```

```
-- 15. TOTAL SUCCESSFUL PAYMENT AMOUNT
```

```
-- =====
```

```
SELECT
    SUM(amount) AS total_successful_amount
FROM Payment
WHERE payment_status = 'Successful';
```

```
-- =====
```

```
-- 16. TOTAL FAILED PAYMENT AMOUNT
```

```
-- =====
```

```
SELECT
    SUM(amount) AS total_failed_amount
FROM Payment
WHERE payment_status = 'Failed';
```

```
-- =====
```

```
-- 17. HIGH-VALUE PAYMENTS ABOVE 10000
```

```
-- =====
```

```
SELECT
    payment_id,
    customer_id,
    order_id,
    payment_mode,
    amount,
    payment_status
FROM Payment
WHERE amount > 10000
ORDER BY amount DESC;
```

```
-- =====
```

```
-- 18. CUSTOMER-WISE PAYMENT REPORT
```

```
-- =====
```

```
SELECT
```

```
    c.customer_id,
```

```
    c.customer_name,
```

```
    COUNT(p.payment_id) AS total_payments,
```

```
    SUM(
```

```
        CASE
```

```
            WHEN p.payment_status = 'Successful'
```

```
            THEN p.amount
```

```
            ELSE 0
```

```
        END
```

```
    ) AS successful_amount
```

```
FROM Customer c
```

```
LEFT JOIN Payment p
```

```
ON c.customer_id = p.customer_id
```

```
GROUP BY c.customer_id, c.customer_name
```

```
ORDER BY successful_amount DESC;
```

```
-- =====
```

```
-- 19. DAILY PAYMENT REPORT
```

```
-- =====
```

```
SELECT
```

```
    payment_date,
```



```
    COUNT(*) AS total_transactions,
    SUM(amount) AS total_amount
FROM Payment
GROUP BY payment_date
ORDER BY payment_date;
```

```
-- =====
```

```
-- 20. COMPLETE PAYMENT TRANSACTION REPORT
```

```
-- =====
```

```
SELECT
    p.payment_id,
    c.customer_name,
    p.order_id,
    p.payment_mode,
    p.payment_date,
    p.amount,
    p.payment_status
FROM Payment p
JOIN Customer c
ON p.customer_id = c.customer_id
ORDER BY p.payment_id;
```